europass

Anastasia Rusakovich

Work permit: France | **Date of birth:** 30/01/1994 | **Place of birth:** Moscow, Russia | **Nationality:** Russian | **Gender:** Female | **Phone:** (+33) 669133518 (Mobile) | **Email address:** anastasiia.rusakovich@gmail.com | **Email address:** anastasia.rusakovich@univ-rennes.fr | **LinkedIn:** anastasia-rusakovich | **Address:** 35000, Rennes, France (Home)

WORK EXPERIENCE

01/10/2023 - 01/11/2026 - RENNES, FRANCE

PHD STUDENT (LONG-READ TRANSCRIPTOMICS & CANCER RESEARCH) INSTITUTE GENETICS & DEVELOPMENT OF RENNES

Bioinformatics:

- Development of *in silico* simulation frameworks for circRNA ground-truth generation (Python, Bash, R)
- Benchmarking of bioinformatics tools using standardised performance metrics
- Oxford Nanopore (ONT) long-read sequencing data analysis
- Differential expression and isoform profiling in cutaneous melanoma (BRAF inhibitor-sensitive and resistant melanoma cell lines)
- Identification circRNA candidates as potential biomarkers and therapeutic targets
- Data analysis on HPC clusters (SLURM, GenOuest)
- Creation of reproducible workflows and version control (Nextflow, GitHub, GitLab)
- Containerisation of tools and environments for community (Conda, Docker, Singularity)

Academic activities:

- Literature review and study design
- Presentation of work at national, international conferences and internal lab seminars
- Publication of findings in peer-reviewed journals

Website: https://igdr.univ-rennes.fr/en

01/04/2019 - 07/11/2023 - DUBNA, RUSSIA

RESEARCH ENGINEER (MOLECULAR BIOLOGY & BIOINFORMATICS) JOINT INSTITUTE FOR NUCLEAR RESEARCH (JINR)

Wet-lab work:

- Preparation of *H. sapiens*, *D. melanogaster*, and *E. coli* samples
- PCR, RT-PCR, iPCR, Long-PCR
- Primers and TaqMan probes design
- Sanger sequencing and analysis
- Sample preparation for NGS (Illumina)
- Construction, cloning, and extraction of plasmids

Bioinformatics:

- Quality control, trimming, mapping, and variant analysis for NGS (Illumina)
- Scripting for data analysis (bash, R, Python)

Academic activities:

- Presentation of work at national, international conferences, science festivals and internal lab seminars
- Supervision of international Masters students
- Literature review and study design
- Publication of national patents

05/09/2017 - 31/03/2019 - DUBNA, RUSSIA

LABORATORY ASSISTANT (CELL CULTURE) JOINT INSTITUTE FOR NUCLEAR RESEARCH, DUBNA (RUSSIA)

Wet-lab work:

- Maintenance of mammalian cells culture (HEK293)

Academic activities:

- Literature review

19/07/2016 - 30/08/2016 - DUBNA, RUSSIA

LABORATORY ASSISTANT (NEUTRON ACTIVATION ANALYSIS) JOINT INSTITUTE FOR NUCLEAR RESEARCH (JINR)

Wet-lab work:

- Plant, moss, and meteorite sample preparation for neutron activation analysis

EDUCATION & TRAINING

01/11/2023 - 01/11/2026 - RENNES, FRANCE

PHD STUDENT- INSTITUTE GENETICS & DEVELOPMENT OF RENNES

Field(s) of study: Bioinformatics, Cancer research, Molecular biology | **Thesis:** Long-read sequencing to investigate the role of circular RNA in melanoma treatment resistance and identify novel biomarkers | **Website:** https://igdr.univ-rennes.fr/en

01/09/2019 - 15/02/2022 - MOSCOW, RUSSIA

MASTER OF SCIENCE- MOSCOW STATE ACADEMY OF VETERINARY MEDICINE AND BIOTECHNOLOGY — MVA BY K.I. SKRYABIN

Field(s) of study: Molecular biology, Biophysics | **Final grade:** Honours degree | **Thesis:** Development of a method for determining the presence of a genetic predisposition to human longevity based on the analysis of FOXO3A gene polymorphism | **Address:** st. Academician Scriabin 23 | **Website:** www.mgavm.ru

01/09/2015 - 26/06/2019 - MOSCOW, RUSSIA

BACHELOR OF SCIENCE- MOSCOW STATE ACADEMY OF VETERINARY MEDICINE AND BIOTECHNOLOGY — MVA BY K.I. SKRYABIN

Field(s) of study: Genetics, Selection | **Final grade:** Honours degree | **Thesis:** Development of a complex system for predicting the risk of IgE-dependent diseases by analyzing the rs2251746 and rs2427837 polymorphisms in the FCER1A gene | **Address:** st. Academician Scriabin 23 | **Website:** www.mgavm.ru

29/08/2023 - 07/10/2023 - MOSCOW, RUSSIA

MACHINE LEARNING IN PYTHON FOR SOLVING BIOINFORMATICS PROBLEMS (ADVANCED TRAINING COURSE)- BLASTIM

Address: Moscow, st. Lenskaya 2/21, office 307 129327 | **Website:** https://blastim.ru/

28/08/2023 - 30/08/2023 - MOSCOW, RUSSIA

SUMMER SCHOOL ON ML IN BIOINFORMATICS- NATIONAL RESEARCH UNIVERSITY HIGHER SCHOOL OF ECONOMICS (HSE)

Website: https://cs.hse.ru/mlbio/2023/

16/05/2023 - 15/06/2023 - MOSCOW, RUSSIA

PYTHON FOR DATA ANALYSIS IN SCIENCE (ADVANCED TRAINING COURSE)- BLASTIM

Address: Moscow, st. Lenskaya 2/21, office 307 129327 | **Website:** https://blastim.ru/

03/06/2023 - 09/06/2023 - MOSCOW, RUSSIA

NGS DATA ANALYSIS (ADVANCED TRAINING COURSE)- INSTITUTE OF SYNTHETIC BIOLOGY

Address: st. Dmitry Ulyanov, house 42 building 1, floor 6, room 1 117218 | **Website:** https://biomedschool.ru/

27/03/2023 - 28/04/2023 - MOSCOW, RUSSIA

STATISTICS, R AND DATA ANALYSIS (ADVANCED TRAINING COURSE)- BLASTIM

Address: Moscow, st. Lenskaya 2/21, office 307 129327 | **Website:** https://blastim.ru/

25/03/2023 - 27/03/2023 - MOSCOW, RUSSIA

PYTHON AND R FOR NGS DATA ANALYSIS (ADVANCED TRAINING COURSE)- INSTITUTE OF SYNTHETIC BIOLOGY

Address: st. Dmitry Ulyanov, house 42 building 1, floor 6, room 1 117218 | **Website:** https://biomedschool.ru/

19/02/2020 - 20/02/2020

SCIENTIFIC PRESENTATIONS (ADVANCED TRAINING COURSE)- SCIENCE BIRDS

Website: https://sciencebirds.de/

PUBLICATIONS

Benchmarking circRNA Detection Tools from Long-Read Sequencing Using Data-Driven and Flexible Simulation Framework 2026

Developed an open-source simulation pipeline and conducted the first performance benchmark of circRNA detection tools for Oxford Nanopore long-read sequencing data.

Anastasia Rusakovich, Sebastien Corre, Edouard Cadieu, Rose-Marie Fraboulet, Victor Le Bars, Marie-Dominique Galibert, Thomas Derrien,

Yuna Blum bioRxiv 2025.04.17.649290; doi: https://doi.org/10.1101/2025.04.17.649290. PCI Genomics

Patent for a method for determining the presence of a genetic predisposition to human longevity 2020

Developed a cost-effective biotechnological method for predicting human longevity genetic predisposition based on the rs2802288 locus.

Patent for a method for predicting the risk of developing human diseases associated with the level of immunoglobulin E (IgE) in humanblood serum 2019

Developed a molecular diagnostic method for predicting risk of IgE-dependent diseases through serum-based genetic analysis.

LANGUAGE SKILLS

Mother tongue(s): **RUSSIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
FRENCH	B1	B1	B1	B1	B1